CITIES AS PORTFOLIOS*

Ailin Zhang†

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Abstract

This paper views cities as portfolios of industries and uses this perspective to evaluate urban performance jointly in terms of wage growth and wage-growth volatility. Using U.S. data from 1990 to 2020, it shows that larger cities systematically achieve more favourable positions in the joint growth–risk space. A symmetric decomposition attributes this efficiency advantage to distinct forces on the growth and risk dimensions. On the growth side, larger cities benefit from both more favourable initial industry composition and stronger reallocation gains, including shifts toward faster-growing industries and wage-tier sorting. On the risk side, their advantage does not reflect safer inherited benchmark exposure, but instead is concentrated in local adjustment margins. To shed light on the mechanisms suggested by the decomposition, the paper exploits U.S. trade liberalization with China as a spatially varying adverse shock. Cities with greater skill portability and a higher prevalence of medium-to-large establishments experience significantly smaller wage losses following the shock, and the inclusion of these two margins substantially attenuates the estimated scale-dependent resilience advantage. Overall, the findings show that differences in urban growth–risk performance reflect the interaction between inherited industrial structure and local adjustment margins.

Keywords: urban growth and risk; wage volatility; industry composition; shock transmission.

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†London School of Economics and Political Science. Email: a.zhang12@lse.ac.uk.

1 INTRODUCTION

Cities differ markedly in how they combine wage growth and wage stability. Some sustain rapid wage growth with little volatility, while others experience large swings with limited long-run gains. These differences become especially salient during major aggregate shocks. When the 2008 financial crisis hit, wages in Austin and Denver dipped only mildly and volatility soon stabilized, whereas in Detroit and Cleveland wages fell sharply and remained unstable long after the recovery. What determines these stark differences in cities' joint growth–risk performance?

Classic urban theories such as Marshallian agglomeration and labour market pooling explain how larger cities may achieve higher productivity and better insulate workers from idiosyncratic firm-level shocks through thicker markets. Yet this emphasis on conventional urban advantages leaves two important gaps. First, it does not explain why wage stability varies widely even among similarly large cities, with some exhibiting persistently high volatility. Second, it has focused primarily on risk pooling at the individual or firm level, with much less attention to aggregate wage risk arising at the city–industry level. Such risk is driven by common sectoral shocks and their interaction with local industrial structure, and cannot be diversified away simply by having more firms. Together, these gaps make it difficult to infer cities' joint growth–risk performance from standard agglomeration mechanisms alone.

This paper studies urban labour markets through a portfolio lens, treating cities as portfolios of industries jointly exposed to common sectoral shocks. In this framework, aggregate wage growth reflects the weighted performance of local industries, while aggregate wage risk reflects the volatility and co-movement structure of the local industry portfolio. This perspective differs from firm-level risk pooling, which primarily mitigates idiosyncratic shocks within firms, by highlighting how common sectoral shocks shape city-level wage dynamics.

The portfolio language does more than relabel urban outcomes. It first provides a disciplined way to compare cities in a joint growth–risk space by measuring how close they lie to an empirical efficient frontier. It then helps explain why cities occupy different positions in that space by separating inherited industrial exposure from subsequent local adjustment. More broadly, it suggests that classic urban questions surrounding specialization versus diversification, structural change, and the attractiveness of locations cannot be evaluated through average growth alone, but must also account for the risks implied by industrial composition and by the scope for adjustment once shocks occur.

The analogy to financial portfolios is informative but not literal, and its limits matter for interpretation. Cities do not deliberately choose or trade their "assets" in the way financial investors do: industrial composition emerges from firms' location decisions and local equilibrium forces rather than portfolio choice. Nor is adjustment frictionless. Reallocating activity across industries requires firm entry and exit, worker mobility, and adjustment under institutional and spatial frictions, so rapid rebalancing is far more limited than in financial markets. Industries are also not standardized assets: the same sector may embody different firm compositions and technologies across cities, and there is no centralized market in which such claims are traded. The paper therefore uses the portfolio lens as a disciplined organizing device rather than a literal asset-market analogy. At the same time, it adopts an industry-based perspective and therefore abstracts from other dimensions of urban performance—most notably amenities and housing markets—which may also shape cities' growth–risk profiles and remain natural extensions for future work.

The first part of the paper maps U.S. cities (CBSAs) into a joint growth–risk space by relating long-run wage growth to wage-growth volatility over 1990–2020, thereby making the urban portfolio perspective operational. This mapping provides a natural way to evaluate urban performance jointly along the growth and risk dimensions. Motivated by the mean–variance framework of Markowitz (1952), I construct an empirical urban efficient frontier and measure each city's position by its distance to this frontier. A clear pattern emerges: larger cities lie systematically closer to the efficient frontier and thus exhibit more favourable joint growth–risk performance.

Mapping cities into growth–risk space is informative about relative performance, but it does not by itself explain why some cities achieve more favourable outcomes than others. To address that question, I develop a symmetric decomposition framework—akin to performance attribution in finance (Brinson et al., 1986)—that decomposes both city growth and city risk into inherited benchmark exposure, benchmark-relative within-industry deviations, and two reallocation margins: shifts toward nationally faster-growing industries and wage-tier sorting across the local wage hierarchy. The decomposition shows that large cities' growth advantage primarily reflects a more favourable initial industry composition and stronger reallocation gains along both margins. Their risk advantage, by contrast, does not arise from safer inherited exposure, but from local adjustment, especially benchmark-relative within-industry deviations and wage-tier sorting. These results are informative about where large cities' growth–risk advantage comes from, but they do not by themselves establish why some cities are better able to reallocate activity and absorb shocks.

The second part of the paper therefore turns from attribution to adjustment mechanisms. While the decomposition treats wage risk symmetrically as a second moment, adverse shocks provide a particularly relevant setting for mechanism analysis because they simultaneously weaken wage growth and worsen cities' joint growth–risk performance, making them especially informative about why cities differ in long-run growth–risk outcomes. Guided by the decomposition, the analysis first develops a simple organizing framework around two broad local adjustment margins—within-industry shock transmission and across-industry labour reallocation. It then studies two candidate city characteristics that may reduce risk under adverse shocks: skill portability, which facilitates employment reallocation across industries, and the prevalence of medium-to-large establishments, which may strengthen within-industry shock buffering. I examine these mechanisms by exploiting U.S. trade liberalization with China, triggered by the granting of PNTR (Permanent Normal Trade Relations), as a spatially varying exogenous shock to local wage dynamics, using city-level exposure measures constructed from Pierce and Schott (2020). A dynamic difference-in-differences approach first shows that cities more exposed to PNTR through their initial industry composition experienced significant wage losses. It then shows that by 2010, an interquartile-range (IQR) increase in skill portability attenuates the wage decline associated with an IQR increase in PNTR exposure by approximately 1.8 log points, offsetting roughly 49% of the baseline aggregate wage loss. A higher pre-shock prevalence of medium-to-large establishments also buffers the shock, preserving about 1.0% of local wages and insulating workers from 27% of the baseline effect. Mediation analysis further indicates that these two channels account for more than 75% of the total scale-dependent resilience. Taken together, the results suggest that inherited portfolio structure shapes cities' baseline exposure to wage risk, while local adjustment capacity governs how shocks are transmitted once they occur.

The paper contributes to several strands of the literature. First, it contributes to the literature on Marshallian agglomeration economies, labour market pooling, and matching externalities (Marshall, 1890; Duranton and Puga, 2004). Existing explanations for the volatility advantage of

large labour markets emphasize more efficient worker–firm matching and lower adjustment costs (Conte et al., 2024; Moretti and Yi, 2024), or the diversification of firms' idiosyncratic productivity shocks (Krugman, 1991; Overman and Puga, 2010; Kikuchi and O'Connor, 2025). Complementing this literature, this paper takes an industry-based perspective and develops a unified framework for jointly analysing wage growth and wage risk, highlighting inherited industrial composition and local adjustment margins as joint determinants of urban growth–risk outcomes. In this respect, it is closely related to Heblich et al. (2025), who shows that early industrial specialization can have persistent adverse effects on city productivity. While that study emphasizes inherited structure and path dependence, this paper shows that, beyond initial composition and diversification, differences in across-industry worker reallocation and within-industry responses to common industry shocks also play a central role in shaping long-run urban performance.

Second, this paper contributes to the literature on the determinants of wage risk. Under incomplete markets and borrowing constraints, uninsurable wage risk plays a central role in shaping individuals' life-cycle savings, consumption, and human capital investment (Levhari and Weiss, 1974; Aiyagari, 1994; Ganong et al., 2025), and can amplify aggregate fluctuations in response to economy-wide shocks (Debortoli and Galí, 2024). Yet the existing literature has focused primarily on wage and earnings risk associated with individual characteristics, such as age and gender (Gottschalk et al., 1994; Gottschalk and Moffitt, 2009; Guvenen et al., 2017), paying much less attention to its geographic dimension. In particular, how heterogeneity in local labour market structure generates systematic, place-based wage volatility remains understudied. Recent urban models typically conceptualize uninsurable income risk as individual-level productivity shocks evolving over the life cycle (Greaney et al., 2025). Even in frameworks that explicitly incorporate location-specific income risk, such risk is usually treated as an exogenous input rather than as an endogenous outcome of local industrial structure and adjustment processes (Ortalo-Magné and Prat, 2016; Hizmo, 2023). This paper instead studies wage risk as a place-based outcome shaped jointly by inherited industrial exposure and local adjustment capacity.

Third, this paper relates to an emerging urban-finance literature that views location choice as an intertemporal investment or self-insurance decision (Ortalo-Magné and Prat, 2016; Morten, 2019; Bilal and Rossi-Hansberg, 2021; Anstreicher, 2024). The closest related study is Bilal and Rossi-Hansberg (2021), who model locations as "assets" that involve current housing costs and future payoffs and whose positions can be adjusted through migration across places. This paper complements that perspective by emphasizing that locations differ not only in expected growth opportunities, but also in the wage risk associated with those opportunities, so that the welfare consequences of location choice depend on both dimensions jointly.

The remainder of the paper is organized as follows. Section 2 maps cities into a growth–risk space, evaluates their growth–risk efficiency, and decomposes wage growth and risk using a symmetric attribution framework. Section 3 presents a simple organizing framework for the local adjustment margins highlighted by the decomposition. Section 4 describes the data. Section 5 uses the China trade shock to examine the mechanisms behind these margins. Section 6 concludes.

2 URBAN GROWTH–RISK EFFICIENCY

2.1 Cities as Portfolios

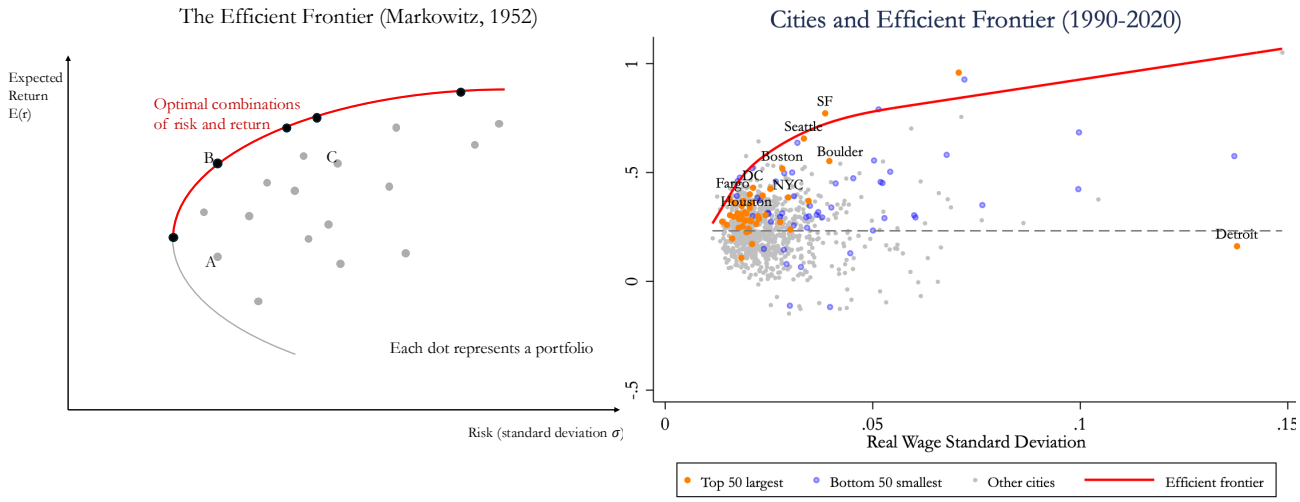


Figure 1: Efficient frontiers in finance and urban world

Figure 1 introduces the central analogy of the paper by mapping both financial portfolios and cities into a common growth–risk space. The left panel illustrates the canonical mean–variance framework in finance (Markowitz, 1952). Each point represents a feasible portfolio constructed from the universe of risky assets, with expected return on the vertical axis and return volatility on the horizontal axis. The efficient frontier traces the upper envelope of the opportunity set: it consists of portfolios that achieve the highest expected return for a given level of risk, or equivalently, the lowest risk for a given level of return. The right panel applies the same mapping to U.S. core-based statistical areas (CBSAs) using data from Quarterly Census of Employment and Wages (QCEW), plotting cumulative real wage growth against the time-series standard deviation of real wage growth over 1990–2020. The urban efficient frontier is constructed empirically by tracing the upper envelope of the city cloud: cities are sorted by risk, record-high growth observations define an outer frontier, this is augmented with a second near-frontier layer for robustness, and a smooth cubic-spline curve is fitted through the resulting hull points, which together define the set of empirically efficient cities.

The analogy becomes natural once cities are viewed as portfolios of industries. Industries correspond to assets, and a city's industrial composition determines its portfolio weights. Aggregate wage growth reflects the weighted average of industry-level growth, while aggregate wage risk depends jointly on within-industry volatility and cross-industry covariance. As in finance, diversification reduces aggregate risk only to the extent that shocks are imperfectly correlated across industries. Cities therefore occupy distinct positions in the growth–risk space, depending on both their exposure to heterogeneous industry assets and the structure of co-movement across industries. As illustrated in Figure 1, cities such as Seattle and Boston lie close to the urban efficient frontier, combining strong long-run growth with moderate risk, whereas cities such as Detroit lie well inside the frontier, exhibiting high wage volatility without commensurate returns. Comparing the 50 largest cities (orange) with the 50 smallest (blue) reveals a clear pattern: large cities lie systematically closer to the efficient frontier, while smaller cities are more dispersed and farther from it.

Importantly, the portfolio analogy is descriptive rather than normative. Cities do not optimally choose or freely rebalance portfolios as financial investors do; industrial composition evolves through firm dynamics, worker mobility, and local adjustment frictions. The value of the analogy lies in elevating wage volatility to a first-order object of comparison and in motivating a coherent way to trace the sources of cities' joint growth–risk performance. Wage volatility is economically salient because it is the risk borne by workers: by locating in a city, they pay housing and mobility costs in exchange for expected wage growth, while remaining exposed to fluctuations in that wage path—much like investors holding a risky asset. Evaluating cities in a joint growth–risk space offers a unified language for comparing urban performance and sets the stage for the decomposition of growth and risk into exposure, reallocation, and idiosyncratic components developed in subsequent sections. See Appendix A for details on the rationale and empirical implementation of the efficient frontier.

2.2 Measuring City Portfolio Efficiency

To move beyond visual comparisons and quantify the growth–risk efficiency of city portfolios, this paper introduces three related distance-based measures, drawing on insights from Kandel and Stambaugh (1995), Li et al. (2003), and Basak et al. (2002). As shown in the left panel of Fig 2, the first is an overall measure—the efficiency gap—defined as the shortest Euclidean distance from a city to the efficient frontier in the normalized growth–risk space, with smaller values indicating higher growth–risk efficiency. Two complementary directional measures characterize where this inefficiency comes from. The growth gap is the vertical distance to the frontier and captures, for a given level of wage risk, how much growth a city forgoes relative to its most efficient counterpart. The risk gap is the horizontal distance and measures, for a given level of wage growth, the excess volatility borne relative to the most efficient benchmark. While these two gaps are mechanically correlated, they emphasize distinct dimensions of inefficiency—foregone growth versus avoidable risk. Compared with conventional portfolio efficiency measures in finance, such as the Sharpe ratio, these distance-based metrics do not require a risk-free asset, making them particularly well suited to the urban context.

A natural question is whether large cities operate more efficiently than small ones when evaluated in a joint growth–risk space. While the agglomeration literature primarily documents scale advantages in levels or growth, it remains largely silent on how growth and volatility jointly scale with city size. In the right panel of Figure 2, I address this question by estimating a simple cross-sectional regression for the period 1990–2020, relating each efficiency measure to initial city size:

$$EffiGap_c = \beta \ln Pop_c + \varepsilon_c. \tag{1}$$

I then plot the estimated coefficient β for each decade and for each of the three efficiency gaps. A negative β indicates that larger cities lie closer to the efficient frontier, while a positive β implies greater size inefficiency. For all three gap measures, the coefficient on initial log population is negative and statistically significant, indicating that larger cities tend to be closer to the growth–risk frontier. See Appendix A for more details.

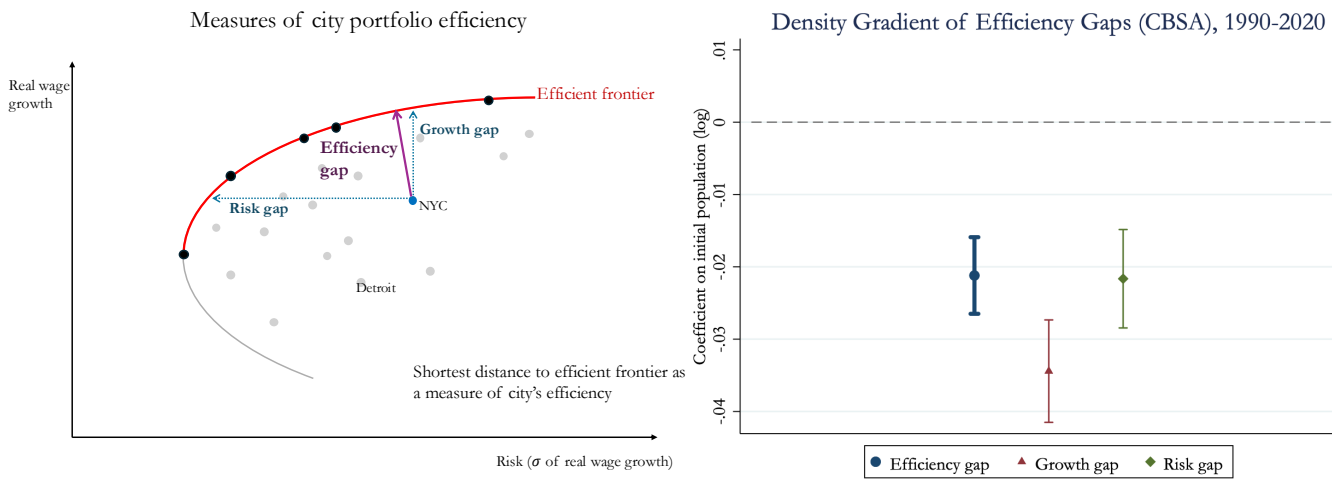


Figure 2: Efficiency measures and density gradients of efficiency gaps. The left panel illustrates three efficiency measures in normalized growth–risk space: the overall efficiency gap (Euclidean distance to the efficient frontier), the growth gap (vertical distance), and the risk gap (horizontal distance). The right panel reports coefficients from cross-sectional regressions of each efficiency measure on initial log city population.

2.3 The Anatomy of Urban Wage Growth and Risk

Mapping cities into a growth–risk space provides a yardstick for evaluating urban performance, much like the mean–variance framework in finance. However, this comparison by itself does not explain why cities occupy different positions relative to the efficient frontier. To address that question, I now turn to a symmetric attribution framework that decomposes realized wage growth and wage risk into inherited industrial exposure and local adjustment margins. Viewing cities as industry portfolios allows us to adapt the performance attribution framework from modern finance (Brinson and Fachler, 1985; Brinson et al., 1986) to the local labour market. While a rich urban literature emphasizes how density economies and thicker labour markets foster urban growth and resilience, these micro-mechanisms rarely provide a unified accounting that separates the mechanical effects of a city's inherited industrial structure from the endogenous responses of its local economy. Unlike standard financial portfolios, where weights reflect invested capital, a city's exposure to its local industries—its "assets"—is weighted by the economic value of its workforce (i.e., payroll shares). Rebalancing this portfolio is therefore not a frictionless transfer of funds, but entails physical—and inherently frictional—reallocation of workers across industries. Extending this framework, I decompose a city's overall performance into four additive components: *benchmark* exposure inherited from initial industry composition, *idiosyncratic* deviations of local industry performance from national trends, and two *reallocation* margins—*growth drift*, capturing shifts toward expanding sectors, and *wage-tier sorting*, capturing worker movements across absolute wage barriers.

2.3.1 Accounting for Wage Growth

We start by unpacking the first moment of city performance: real wage growth. Let $s_{i,c,t}$ denote the employment share of industry $i \in J$ in city c at time t relative to total local employment.[1] Let $w_{i,c,t}$ denote the absolute wage level of industry i. We define the payroll share as $\tilde{s}_{i,c,t} = s_{i,c,t} w_{i,c,t} / W_{c,t}$ and the relative wage as $\tilde{R}_{i,c,t} = w_{i,c,t}/W_{c,t}$, where $W_{c,t} = \sum_i s_{i,c,t} w_{i,c,t}$ is the city's average wage. By decomposing the total change in the city's average wage, the expected real wage growth $G_{c,t}$ of a city portfolio from $t-1$ to t can be linearly approximated as:[2]

$$G_{c,t} \approx \underbrace{\sum_i \tilde{s}_{i,c,t-1}\, g_{i,c,t}}_{\text{Within-industry growth}} + \underbrace{\sum_i \Delta s_{i,c,t}\, \tilde{R}_{i,c,t-1}}_{\text{Wage-tier sorting}} \tag{2}$$

where $g_{i,c,t}$ is the wage growth of industry i, which we decompose into a national common component and a city-industry idiosyncratic component ($g_{i,c,t} = g^N_{i,t} + \tilde{g}_{i,c,t}$). Economically, the first term captures the wage growth *within* incumbent industries weighted by their initial economic value share, while the second term captures the gain from moving workers across industries with different absolute wage levels—a dynamic we term *wage-tier sorting*. Crucially, this two-term structure highlights a fundamental departure from standard financial models, in which the second term is

[1] In the baseline analysis, we focus on the composition of the employed workforce. In Appendix [X], we relax this by anchoring shares to the total local labour force, thereby explicitly internalising *employment risk*. In this expanded framework, any transition into unemployment is mathematically equivalent to an "asset" value vanishing to zero. Our results are robust to this alternative specification.

[2] Algebraically, the exact growth rate is $G_{c,t} = (\sum_i s_{i,c,t} w_{i,c,t} - W_{c,t-1})/W_{c,t-1}$. Using the discrete product rule, this expands to $\sum_i s_{i,c,t-1}(\Delta w_{i,c,t}/W_{c,t-1}) + \sum_i \Delta s_{i,c,t}(w_{i,c,t-1}/W_{c,t-1}) + \epsilon$, where $\epsilon = \sum_i \Delta s_{i,c,t} \Delta w_{i,c,t}/W_{c,t-1}$ is a negligible second-order interaction term. Substituting sectoral wage growth $g_{i,c,t} = \Delta w_{i,c,t}/w_{i,c,t-1}$, payroll share $\tilde{s}_{i,c,t-1}$, and wage premium $\tilde{R}_{i,c,t-1}$ yields this approximation.

mechanically zero. In a financial portfolio, instantaneous rebalancing yields no immediate return because it is governed by a *value-conserving* self-financing constraint—buying one asset requires selling another of equal market value. In contrast, labour mobility is *headcount-conserving*: as workers transition across absolute wage barriers, the city's physical headcount remains constant, but the total monetary value of its portfolio mechanically expands or contracts. Consequently, the net effect of this wage-tier sorting can be either positive or negative. It yields a structural premium when workers successfully navigate adjustment frictions to climb into higher-paying sectors. Conversely, it imposes a structural penalty when secular forces, such as job polarization or displacement shocks, compel workers to downgrade into lower-yielding jobs.

We now turn to the evolution of city portfolios over time. Summing the single-period approximations over horizon T and adapting the Brinson (Brinson et al., 1986) framework, we decompose cumulative wage growth into four constituent parts. Crucially, by further unpacking the within-industry component, this formulation yields two distinct *reallocation* margins: a financial-style *growth drift* effect driven by shifting value weights, and a labour-specific *wage-tier sorting* effect driven by the physical mobility of workers:[3]

$$G_{c,T} \approx \underbrace{\sum_{t=1}^{T}\sum_{i} \tilde{s}_{i,c,0}\, g^{N}_{i,t}}_{\text{Benchmark } (\sum r^{B}_{c,t})} + \underbrace{\sum_{t=1}^{T}\sum_{i} \tilde{s}_{i,c,t-1}\, \tilde{g}_{i,c,t}}_{\text{Idiosyncratic } (\sum r^{I}_{c,t})} + \underbrace{\underbrace{\sum_{t=1}^{T}\sum_{i} (\tilde{s}_{i,c,t-1} - \tilde{s}_{i,c,0})\, g^{N}_{i,t}}_{\text{Growth drift } (\sum r^{C}_{c,t})} + \underbrace{\sum_{t=1}^{T}\sum_{i} \Delta s_{i,c,t}\, \tilde{R}_{i,c,t-1}}_{\text{Wage-tier sorting } (\sum r^{S}_{c,t})}}_{\text{Reallocation gain}} \tag{3}$$

The first term, *benchmark growth* (with single-period return $r^{B}_{c,t} = \sum_i \tilde{s}_{i,c,0} g^{N}_{i,t}$), captures the passive wage-growth path inherited from a city's initial payroll structure: it is the growth a city would experience if it retained its baseline industry composition while each sector grew at its national rate. In this sense, it corresponds to the passive return in portfolio attribution, so cities such as San Jose or San Francisco, with large initial exposure to nationally fast-growing technology and professional service sectors, exhibit high benchmark growth even absent local adjustment. The second term, *idiosyncratic growth* ($r^{I}_{c,t} = \sum_i \tilde{s}_{i,c,t-1}\tilde{g}_{i,c,t}$), captures within-industry performance relative to national benchmarks, analogous to a selection effect in portfolio attribution, and is therefore a priori ambiguous in sign: it may reflect local agglomeration-related outperformance, but it may also capture spatial convergence or other city-specific deviations from national industry trajectories. The remaining *reallocation gain* contains two distinct active margins. The *growth drift* term ($r^{C}_{c,t} = \sum_i (\tilde{s}_{i,c,t-1} - \tilde{s}_{i,c,0}) g^{N}_{i,t}$) measures the return from shifting economic weight toward sectors that subsequently experience stronger national growth, much like active portfolio reallocation in finance; but unlike in financial markets, such shifts are difficult because future national growth is only imperfectly foreseeable and labour reallocation is constrained by skill specificity and adjustment frictions. Thus, manufacturing-intensive cities such as Detroit may remain locked into declining activities, whereas cities such as Austin may pivot more quickly toward rising sectors. Finally, the *wage-tier sorting* term ($r^{S}_{c,t} = \sum_i \Delta s_{i,c,t} \tilde{R}_{i,c,t-1}$) has no direct counterpart in standard finance: it captures the income gain or loss generated by actual worker movements across sectors with different absolute wage levels, for example when displaced manufacturing workers are reabsorbed into higher-paying professional services rather than lower-wage retail jobs, and thus reflects a city's capacity to absorb

[3] Our approach adapts the Brinson framework in three ways: we use initial payroll shares as the passive benchmark, evaluate idiosyncratic performance using dynamic contemporaneous weights, and break the value-conserving constraint by explicitly introducing the wage-tier sorting margin.

labour into better-paying activities rather than lower-wage fallback sectors.

Applying the four-part accounting framework to the data, Figure 3 identifies the sources of the large-city real wage premium over 1990–2020. Panel (a) shows a sizable long-run divergence: cities with a 1990 labour force above 0.5 million accumulated about 10 log points more real wage growth than smaller cities over the sample period. Panels (b) and (c), which implement the decomposition at the 4-digit NAICS level, show that the *benchmark* component accounts for most cumulative growth in both city groups. Inherited industry composition therefore remains the main driver of long-run wage growth, consistent with a portfolio-attribution interpretation in which initial exposure determines the baseline trajectory (Brinson et al., 1986). At the same time, this similarity in the benchmark component masks an important difference in the active reallocation margins, particularly *wage-tier sorting*. Small cities experience a large and persistent sorting loss, with cumulative wage growth reduced by nearly 7 log points as workers are reallocated toward lower-wage sectors. Large cities, by contrast, offset comparable early sorting losses over time, so that the sorting component is close to zero by the end of the sample. The *growth drift* margin is positive but quantitatively modest, and the *idiosyncratic* component remains small in both groups. Overall, the results indicate that the large-city growth premium reflects not only more favorable initial industry composition, but also a greater ability to avoid persistent losses from downward labour reallocation.

Panel (d) plots the cumulative Large-minus-Small gap for each component at the 4-digit level. The resulting divergence is not accounted for by the *benchmark* component alone. Both active reallocation margins also contribute: large cities exhibit a steadily widening gap in *wage-tier sorting* and a persistent premium in *growth drift*. By contrast, the *idiosyncratic* gap is strongly time-varying. After being positive early in the sample, it turns negative from the mid-2000s onward, consistent with a form of spatial mean reversion in which smaller cities experience faster within-industry wage growth. Taken together, these patterns indicate that large cities preserve their aggregate wage-growth advantage not only through more favourable initial industry composition, but also through stronger labour reallocation across sectors. Figure 4 shows that these reallocation differences reflect a highly uneven sectoral transformation.[4] Panel A shows that the *growth drift* premium is driven almost entirely by large cities shifting toward nationally expanding *modern tradables*. Panel B shows a polarized pattern in *wage-tier sorting*: large cities incur a sizeable sorting loss from the secular decline of historically middle-wage *traditional tradables*, but offset this loss through positive sorting into *modern tradables*. This pattern is consistent with greater capacity of large cities to reallocate displaced labour toward more productive emerging sectors, a margin that appears weaker in smaller cities.

The importance of the fine-grained decomposition is apparent in Panel (e), which repeats the exercise at the broader 2-digit NAICS level. At this level of aggregation, the sorting gap is substantially compressed, with the *wage-tier sorting* gap turning negative in the 2010s, while the *benchmark* advantage is also attenuated. Both margins instead load into a much larger *idiosyncratic* gap. Comparing Panels (d) and (e) therefore suggests that what appears as within-industry outperformance in coarse sectoral data partly reflects micro-level composition. Large cities do not necessarily generate faster wage growth within narrowly defined jobs; rather, they inherit more favourable 4-digit

[4] We extend Mian and Sufi (2014) to a four-sector taxonomy: (i) *traditional tradables* (primarily manufacturing); (ii) *local non-tradables* (retail and hospitality); (iii) *modern tradables* (knowledge-intensive services with Geographic Herfindahl Index (GHI) ≥ 0.85); and (iv) *others* (construction, dispersed services with GHI < 0.85, and residual sectors). To address QCEW suppression, masked sub-sectors are allocated proportionally based on the observed employment distribution within their parent industries, while fully unclassified records are conservatively assigned to *others*.

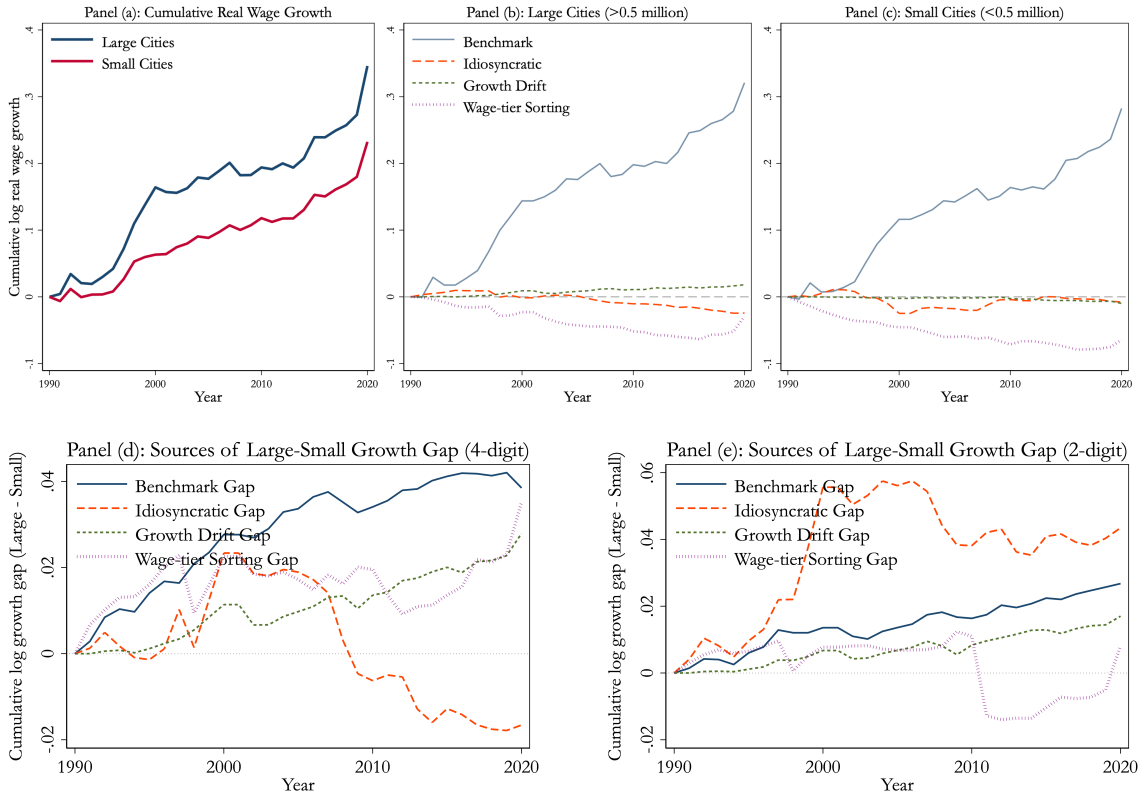


Figure 3: **Decomposition of Urban Wage Growth: Mechanisms and the Role of Granularity.** Panel (a) presents the realized cumulative log real wage growth from 1990 to 2020 for large versus small cities. Panels (b) and (c) decompose this growth into the four additive components defined in Equation 3 at the 4-digit NAICS level: the national benchmark, city-specific idiosyncratic growth, and two reallocation terms (growth drift and wage-tier sorting). Panels (d) and (e) present the decomposition of the cumulative Large–Small growth gap using industry data at the 4-digit and 2-digit NAICS levels, respectively. Large cities ($N = 41$) are CBSAs with a 1990 labor force exceeding 0.5 million; small cities ($N = 868$) are those below this threshold. The analysis is restricted to the private sector. Data source: QCEW, adjusted by CPI-U.

sub-sectors and are better able to reallocate workers toward them, patterns that are obscured at the 2-digit level.[5] See Appendix Section B.1 for additional details and robustness results.

[5]To address mechanical correlation, benchmark growth is computed using a hierarchical leave-one-out (LOO) approach. Specifically, when constructing national industry growth for city c, we exclude city c itself from the national aggregate. This ensures that the benchmark rates are not mechanically influenced by local contemporaneous performance, thereby preserving the interpretation of the benchmark and idiosyncratic components.

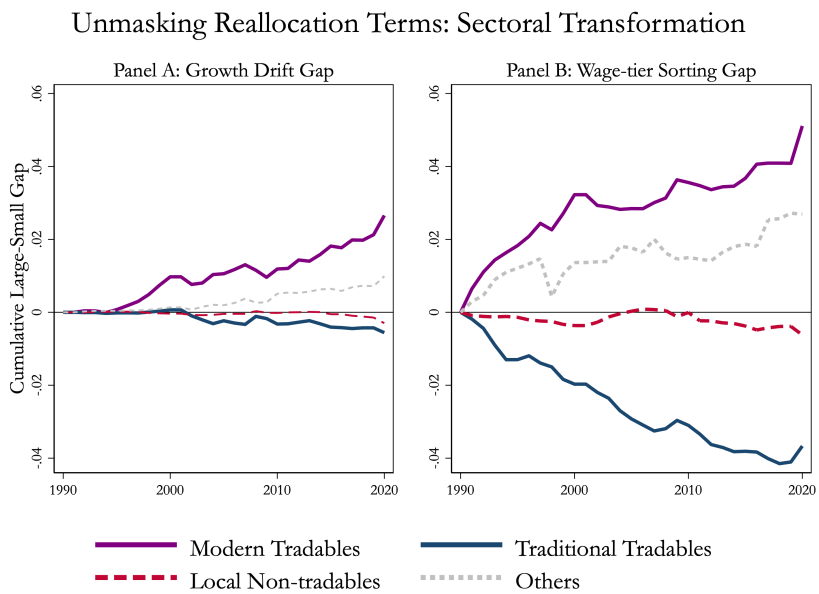


Figure 4: **Sectoral Decomposition of Reallocation Premiums.** Cumulative large–small city gaps in *growth drift* (Panel A) and *wage-tier sorting* (Panel B). The four-sector taxonomy adjusts Mian and Sufi (2014): (i) *traditional tradables* (manufacturing); (ii) *local non-tradables* (retail/hospitality); (iii) *modern tradables* (others with GHI $\geq$ 0.85); (iv) *others* (construction, healthcare, education, and residuals). Suppressed industries are proportionally allocated. Large cities ($N = 41$) are CBSAs with a 1990 labor force exceeding 0.5 million; small cities ($N = 868$) are those below this threshold. The analysis is restricted to the private sector. Data source: QCEW, adjusted by CPI-U.

2.3.2 Accounting for Wage Growth Volatility

We next turn to the second moment of city performance: wage-growth risk, which we measure over horizon T as the time-series standard deviation of annual real wage growth, $\sigma_c \equiv \text{StdDev}_T(G_{c,t})$. As shown in Equation 3, annual city wage growth is additively decomposed as

$$G_{c,t} \approx r^B_{c,t} + r^I_{c,t} + r^C_{c,t} + r^S_{c,t}. \tag{4}$$

This additivity implies that aggregate wage-growth risk can be attributed to the same four structural margins:

$$\sigma_c \approx S^B_c + S^I_c + S^C_c + S^S_c, \qquad S^k_c \equiv \frac{\text{Cov}_T(r^k_{c,t}, G_{c,t})}{\sigma_c} = \sigma_k \rho_{k,G}, \quad k \in \{B, I, C, S\}, \tag{5}$$

where $\rho_{k,G}$ denotes the correlation between margin k and aggregate city wage growth and $\sigma_k \equiv \text{StdDev}_T(r^k_{c,t})$. A margin therefore contributes more to aggregate wage risk when it is both more volatile in its own right and more positively aligned with the overall wage cycle. This formulation preserves symmetry with the growth accounting: the same four objects used to decompose mean wage growth also decompose aggregate wage volatility.

It is important, however, to be precise about what Equation 5 identifies. The objects S^k_c do not isolate the primitive variance of each margin in separation from all other terms, nor do they reveal a one-to-one mapping from each risk contribution to a unique underlying mechanism. Rather, they

summarize the *overall contribution* of each structural margin to aggregate city wage-growth volatility. Under this interpretation, *benchmark risk* captures the contribution of inherited exposure to nationally fluctuating industries. *Idiosyncratic risk* captures the contribution of benchmark-relative local deviations, and therefore points to heterogeneity in within-industry responses to common industrial shocks. The remaining two margins capture the risk implications of reallocation, but along distinct dimensions. *Growth drift risk* reflects the volatility associated with shifting value-weighted exposure toward industries with different national growth paths over time, whereas *wage-tier sorting risk* captures the volatility associated with worker movements across industries with different absolute wage levels in the local wage hierarchy.

Applying this attribution to the data reveals that the lower wage-growth volatility of large cities reflects a markedly different *risk composition*, rather than a uniform reduction in all margins. Figure 5 shows that the *benchmark* contribution is larger in large cities, indicating that their lower overall volatility does not arise from systematically safer inherited exposure to nationally fluctuating industries. Small cities nevertheless exhibit substantially higher total volatility over 1990–2020, and most of the large-city risk advantage is accounted for by their lower *idiosyncratic* margin, with *wage-tier sorting* contributing an additional, though smaller, share. By contrast, the *growth drift* margin contributes little to total risk in either group. Overall, the figure indicates that the risk advantage of large cities is concentrated primarily in the *idiosyncratic* and *wage-tier sorting* margins, rather than in inherited exposure. This pattern points to the importance of benchmark-relative within-industry deviations and across-industry labour reallocation in understanding large cities' superior growth–risk performance. See Appendix Section B.2 for more details.

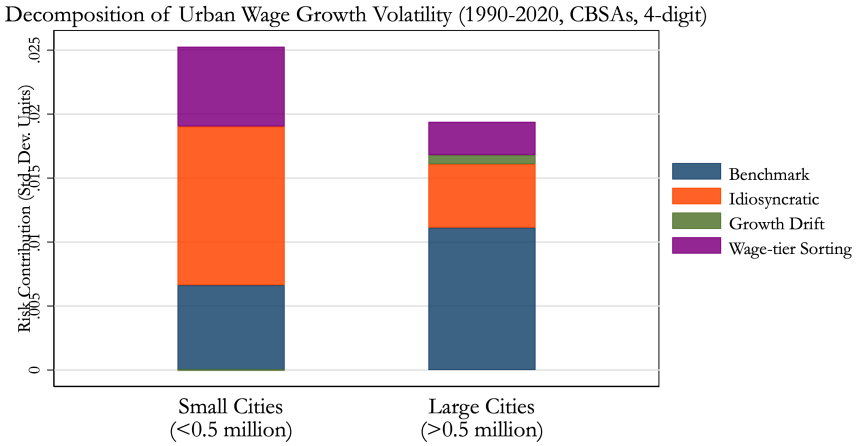


Figure 5: **Decomposition of Urban Wage Growth Volatility.** The figure displays the absolute risk contributions of the four main structural margins (*benchmark*, *idiosyncratic*, *growth drift*, and *wage-tier sorting*) to total urban wage growth volatility from 1990 to 2020 using Equation 5. The computation of each margin follows Equation 3. Large cities ($N = 41$) are CBSAs with a 1990 labour force exceeding 0.5 million; small cities ($N = 868$) are those below 0.5 million. The analysis focuses on private sector in QCEW data.

2.3.3 Implications

Taken together, the decomposition shows that cities' positions in growth–risk space are shaped by two forces. The first is inherited exposure: initial industry composition largely determines the

benchmark path of wage growth and the baseline exposure to nationally fluctuating industries. The second is local adjustment: large cities distinguish themselves not because they inherit safer benchmark exposure, but because their outcomes differ along margins associated with reallocation and benchmark-relative local performance. On the growth side, large cities achieve stronger reallocation gains. On the risk side, their volatility advantage is concentrated in the idiosyncratic and wage-tier sorting margins, rather than in the benchmark component.

These patterns identify where the large-city growth–risk advantage resides, but not the unique mechanisms that generate it. The idiosyncratic margin captures benchmark-relative within-industry deviations. Such deviations may reflect several sources of local heterogeneity, including firm-level granularity, establishment composition, local demand conditions, and differential transmission of common industry shocks. The wage-tier sorting margin is more directly tied to worker-side adjustment: it captures whether labour reallocation moves workers toward better-paying activities or instead pushes them toward lower-wage fallback sectors. Thus, the decomposition points to two broad adjustment domains—within-industry shock transmission and across-industry labour reallocation—without by itself establishing the underlying causal mechanisms.

This motivates the next step of the analysis: probing the local adjustment mechanisms behind large cities' growth–risk advantage. A city's long-run position in growth–risk space reflects how it responds to repeated positive and negative industry shocks. Because the risk decomposition treats wage volatility symmetrically as a second moment, it does not distinguish between upside and downside fluctuations. Adverse shocks provide a clearer setting for mechanism analysis because they directly worsen growth–risk performance by lowering wage growth and revealing differences in local shock transmission. Section 5 therefore exploits U.S. trade liberalization with China as an exogenous adverse shock to local labour markets. The analysis focuses on the role of two local characteristics motivated by the decomposition: skill portability, which facilitates across-industry labour reallocation, and the prevalence of medium-to-large establishments, which may shape within-industry shock transmission. The next section develops a heuristic framework that helps organize the intuition behind these candidate mechanisms before turning to the event-study evidence.

3 A SIMPLE ORGANIZING FRAMEWORK

To organize the mechanism discussion and motivate the event study, I develop a parsimonious framework in which cities are portfolios of industries exposed to common industry shocks. The framework is not meant to exhaust all determinants of urban performance, but instead isolates two local adjustment margins that affect growth and risk suggested by the decomposition: within-industry shock transmission and across-industry labour reallocation.

Environment. Time is discrete, $t = 0, 1, 2, \ldots$. A city c enters period t with industry employment shares

$$s_{c,t-1} \equiv \{s_{ic,t-1}\}_{i=1}^{I}, \qquad \sum_{i=1}^{I} s_{ic,t-1} = 1,$$

where $i \in \{1, \ldots, I\}$ indexes industries. Let

$$W_{c,t-1} = \sum_{i=1}^{I} s_{ic,t-1} w_{ic,t-1}$$

denote the city-wide average wage, and define the lagged relative wage ratio

$$\tilde{R}_{ic,t-1} \equiv \frac{w_{ic,t-1}}{W_{c,t-1}}.$$

Lagged payroll shares are then

$$\tilde{s}_{ic,t-1} = s_{ic,t-1} \tilde{R}_{ic,t-1}.$$

Thus, a city's effective portfolio reflects both its employment composition and its wage hierarchy.

National industry growth and local shock absorption. Let national wage growth in industry i be

$$g_{it}^N = \mu_i + (1 - \Phi_i^+)\varepsilon_{it}^+ - (1 - \Phi_i^-)\varepsilon_{it}^-, \tag{6}$$

where μ_i is the industry's long-run national growth rate, $\varepsilon_t = (\varepsilon_{1t}, \ldots, \varepsilon_{It})'$ is a vector of common industry shocks with

$$\mathbb{E}[\varepsilon_t] = 0, \qquad \mathrm{Var}(\varepsilon_t) = \Sigma^N,$$

and

$$\varepsilon_{it}^+ \equiv \max\{\varepsilon_{it}, 0\}, \qquad \varepsilon_{it}^- \equiv \max\{-\varepsilon_{it}, 0\}.$$

The parameters Φ_i^+ and Φ_i^- summarize the extent to which positive and negative industry-i shocks are absorbed in the observed national benchmark. A higher value implies weaker pass-through of the corresponding shock into national wage growth.

We summarize city c's within-industry shock absorption by two reduced-form parameters, ϕ_c^+ and ϕ_c^-, governing the local absorption of positive and negative industry shocks. Higher values imply stronger local buffering and therefore weaker pass-through into local wages. To keep the framework parsimonious, these parameters are defined at the city level rather than allowed to vary across city–industry cells. They may reflect local features that shape how industry shocks are transmitted into wages, such as establishment scale, local policy environments, or other dimensions of organizational resilience. The framework therefore does not impose a specific micro-foundation.[6] Local wage growth in industry i is then

$$g_{ic,t} = \mu_i + (1 - \phi_c^+)\varepsilon_{it}^+ - (1 - \phi_c^-)\varepsilon_{it}^-. \tag{7}$$

Equivalently,

$$g_{ic,t} = g_{it}^N + \tilde{g}_{ic,t},$$

where

$$\tilde{g}_{ic,t} = (\Phi_i^+ - \phi_c^+)\varepsilon_{it}^+ - (\Phi_i^- - \phi_c^-)\varepsilon_{it}^-.$$

Thus, benchmark-relative local deviations capture how local wage responses to common industry shocks differ from the observed national benchmark, allowing upside and downside absorption to differ. Under an adverse industry shock, a city with higher downside absorption ϕ_c^- experiences a

[6] For example, Appendix C provides a simple illustration of how a larger average establishment scale may strengthen local shock-buffering capacity. This example is heuristic rather than exclusive.

smaller wage decline than the national benchmark if $\phi_c^- > \Phi_i^-$, implying a more positive benchmark-relative deviation $\tilde{g}_{ic,t}$ in bad times. This improves relative performance under adverse shocks. By contrast, the implication for the volatility of the idiosyncratic margin is generally ambiguous. Because the idiosyncratic component is defined relative to the national benchmark, its volatility is not, by itself, a clean measure of local shock-buffering capacity: it depends both on local buffering relative to the benchmark transmission parameters Φ_i^+ and Φ_i^- and on how benchmark-relative deviations evolve across positive and negative shocks. For example, a city may attenuate adverse shocks more strongly in absolute terms and therefore exhibit smoother local wage growth, yet still display sizable benchmark-relative deviations if its transmission differs systematically from the national benchmark. More broadly, the idiosyncratic margin is best interpreted as indicating that cities may differ in within-industry transmission of common industry shocks, rather than as uniquely identifying the source of those differences.

Worker-side reallocation. At the beginning of period t, shocks are realized and workers reallocate across industries. Let

$$\Delta s_{ic,t} \equiv s_{ic,t} - s_{ic,t-1}$$

denote the change in industry i's employment share in city c. The period-t equilibrium wage in industry i is

$$w_{ic,t} = w_{ic,t-1}(1 + g_{ic,t}) - \eta_c s_{ic,t}, \tag{8}$$

where $\eta_c > 0$ governs the steepness of sectoral labour supply within the city. A lower η_c corresponds to a thicker labour market, so that inflows depress wages less.

Workers face mobility frictions across industries. Let $d_{ij} \geq 0$ denote the skill distance between origin industry j and destination industry i, and let $\mathcal{I}_c$ denote the set of industries active in city c. We specify the conditional transition probability

$$\pi_{i|j,c,t} = \frac{\exp\{w_{ic,t} - d_{ij}\}}{\sum_{k \in \mathcal{I}_c} \exp\{w_{kc,t} - d_{kj}\}}, \qquad i \in \mathcal{I}_c, \tag{9}$$

so that workers move more readily toward higher-wage and more skill-proximate destinations. Employment shares evolve according to

$$s_{ic,t} = \sum_{j=1}^{I} s_{jc,t-1} \pi_{i|j,c,t}. \tag{10}$$

This block captures worker-side adjustment capacity. While labour-market thickness η_c and the breadth of the destination set $\mathcal{I}_c$ also matter, the key friction is the skill distance d_{ij} between origin and destination sectors. The model implies that lowering skill frictions to sufficiently high-wage destinations raises the choice-weighted lagged destination wage of reallocated workers and thereby strengthens the wage-tier sorting margin. More broadly, the value of industry breadth lies not in the sheer number of sectors per se, but in the availability of high-wage and skill-proximate destinations. Conversely, when skill portability is weak—that is, when skill distances are high—or when the expansion of the destination set mainly adds relatively low-wage fallback sectors rather than better-paying opportunities, labour reallocation is more likely to generate downward sorting losses. To the extent that larger cities offer denser sets of high-wage and skill-proximate destinations, the model

predicts a worker-side adjustment advantage for larger cities. Appendix C.2 derives these results formally.

City wage growth. Aggregate wage growth in city c is

$$G_{c,t} = \underbrace{\sum_{i=1}^{I} \tilde{s}_{ic,t-1} g_{ic,t}}_{\text{within-industry wage growth}} + \underbrace{\sum_{i=1}^{I} \Delta s_{ic,t} \tilde{R}_{ic,t-1}}_{\text{wage-tier sorting}} . \tag{11}$$

Substituting (7) into (11), and adding and subtracting the benchmark return based on the city's initial payroll structure, yields

$$G_{c,t} = \underbrace{\sum_{i} \tilde{s}_{ic,0} g_{it}^{N}}_{\text{Benchmark}} + \underbrace{\sum_{i} \tilde{s}_{ic,t-1} \tilde{g}_{ic,t}}_{\text{Idiosyncratic}} + \underbrace{\sum_{i} (\tilde{s}_{ic,t-1} - \tilde{s}_{ic,0}) g_{it}^{N}}_{\text{Growth drift}} + \underbrace{\sum_{i} \Delta s_{ic,t} \tilde{R}_{ic,t-1}}_{\text{Wage-tier sorting}} . \tag{12}$$

The first term is the passive return inherited from the city's initial payroll structure. The second captures benchmark-relative local deviations within industries. The fourth captures the contemporaneous wage effect of workers moving across the local wage hierarchy. The third then captures the delayed portfolio consequence of those past reallocations: by changing payroll weights over time, earlier worker movements alter how much of subsequent national industry growth the city is exposed to. In this sense, wage-tier sorting is the short-run return to labour mobility, whereas growth drift is its intertemporal portfolio consequence.

City wage risk. Because aggregate wage growth is the sum of the same four components,

$$G_{c,t} = r_{c,t}^{B} + r_{c,t}^{I} + r_{c,t}^{C} + r_{c,t}^{S},$$

city risk is shaped by the volatility and comovement of these margins. The idiosyncratic margin records benchmark-relative local deviations within industries and therefore signals heterogeneity in within-industry responses to common industry shocks, but it is not a monotone measure of local shock-buffering capacity. The wage-tier sorting margin provides a more direct window into worker-side adjustment:

$$r_{c,t}^{S} = \sum_{i} \Delta s_{ic,t} \tilde{R}_{ic,t-1}.$$

It is less volatile when labour reallocation is diffuse across industries and less tilted toward extreme positions in the local wage hierarchy. The framework is therefore intended to clarify the economic sources behind the empirical margins, rather than to imply a one-to-one mapping from each risk component to a unique mechanism.

Interpretation. The framework organizes the empirical analysis by distinguishing inherited benchmark exposure from local adjustment margins. In the model, cities differ not only in their initial industry portfolios, but also in how strongly common industry shocks are transmitted into local wages and how effectively workers can be reallocated across industries once shocks occur. The decomposition therefore identifies which margins matter beyond initial industry structure, while the

subsequent empirical analysis uses separate empirical designs to shed light on the candidate mechanisms underlying those margins. At a parsimonious level, the key reduced-form objects are the local shock-transmission parameters ϕ_c^+ and ϕ_c^-, the skill-distance matrix d_{ij}, and the breadth of the local destination set $\mathcal{I}_c$. Together, these objects summarize the city characteristics that govern within-industry shock transmission and worker-side adjustment in the model. A limitation of the framework is that it adopts an industry-level perspective on common industry shocks and local adjustment, and therefore abstracts from firm-level idiosyncratic shocks.

4 DATA

This section describes the data used in the analysis. It first introduces the wage and employment data used to construct city-level measures of wage growth and time-series wage risk underlying the portfolio analogy and growth–risk decomposition. It then presents the data used to construct two predetermined city characteristics capturing local adjustment capacity: skill portability and the prevalence of medium-to-large establishments. Finally, it describes the data used in the event-study analysis of the China trade liberalization shock. See Appendix D for more details.

4.1 Wage and employment

The data on annual wages and employment counts from 1990 to 2020 are drawn from the publicly-available Bureau of labour Statistics Quarterly Census of Employment and Wages (BLS QCEW), which provides quarterly county–industry administrative records based on employer filings under state unemployment insurance (UI) programs and is aggregated to 2013 CBSAs. This paper focuses on the private sector, where wages and employment are primarily determined by market forces rather than institutional pay-setting. Our baseline analysis uses NAICS 4-digit industries, comprising 305 unique industries observed over the sample period. Wages are measured as QCEW average annual pay, defined as total annual payroll divided by average monthly employment, which provides a comprehensive, administratively recorded measure of per-worker labour income, including salaries, bonuses, tips, and other cash compensation, though it may also reflect changes in hours and workforce composition. Quarterly observations are aggregated to annual values and deflated using the BLS CPI-U. At the city–industry level, real wage growth is computed as $g_{ict} = \ln w_{ict} - \ln w_{ic,t-1}$. City-level average wages are constructed as $W_{ct} = \sum_i s_{ict} w_{ict}$, where s_{ict} is industry i's employment share in city c. City-level wage growth is then $G_{ct} = \ln W_{ct} - \ln W_{c,t-1}$, and city-level wage-growth risk over period p is measured as $\sigma_{cp} = \mathrm{sd}_{t \in p}(G_{ct})$.

A key challenge in the QCEW is confidentiality-induced suppression: the BLS masks employment and wage counts in industry–county cells with few establishments, causing disclosed 4-digit industries to fall short of reported county-level totals. We address this using a recursive residual-attribution procedure: whenever a parent aggregate (e.g., 3-digit NAICS) exceeds the sum of its disclosed subsectors (e.g., 4-digit), we assign the residual to a *synthetic suppressed sector* within that parent. This ensures that 100% of city-level private-sector employment and labour income are retained while preserving industry identity at the highest feasible resolution. For the post-2004 period, when establishment counts for suppressed cells are available, we further validate this approach using the granular imputation procedure of Autor et al. (2024), detailed in Appendix Section D.1.

4.2 Skill portability

Skill portability is treated as an initial city characteristic measured prior to the China trade shock. We construct skill portability using individual-level data from the 1990 5% IPUMS sample. The sample is restricted to employed, private-sector wage and salary workers who worked at least 40 weeks in the previous year and usually worked at least 35 hours per week, and their industry and occupation information is used to construct baseline city-level measures. [7]

A growing literature emphasizes that human capital specificity is primarily skill- rather than industry-based (Poletaev and Robinson, 2008), implying that cities' adjustment to industry shocks depends on the extent to which existing skills can be redeployed across industries. Most existing measures of skill portability focus on individual outcomes, such as task similarity or wage changes following job switches (Gathmann and Schönberg, 2010; Yi et al., 2017; Manning and Aguirre, 2025). While informative about individual mobility, these approaches do not capture cross-city differences in the aggregate redeployability of skills or cities' capacity to reallocate workers across industries in response to large industry shocks.

To capture this spatial difference in labour mobility, this paper develops a novel measure of a city's *skill portability* as the expected probability that a worker can reallocate to another local industry while remaining in the same occupation.[8] This measure builds on the assumption that each occupation represents a bundle of occupation-specific skills that can be deployed across multiple local industries. Consider a worker employed in occupation–industry cell (o, j) in city c. Conditional on staying in occupation o, the probability of being re-employed in a different industry $j' \neq j$ equals $(1 - s^c_{j|o})$, where $s^c_{j|o} \equiv E^c_{oj}/E^c_o$ denotes the employment share of industry j within occupation o in city c. Aggregating over all occupation–industry cells yields the city-level average reallocation probability:

$$P_c = \sum_{o,j} s^c_{oj}\big(1 - s^c_{j|o}\big) = \sum_o s^c_o D^c_o, \qquad D^c_o = 1 - \sum_j \big(s^c_{j|o}\big)^2, \tag{13}$$

where $s^c_{oj} = E^c_{oj}/L^c$ is the fraction of total city employment L^c accounted for by occupation–industry cell (o, j), and $s^c_o = E^c_o/L^c = \sum_j E^c_{oj}/L^c$ denotes the share of occupation o in the city's overall employment. The first equality follows from the law of total probability: the overall reallocation probability equals the weighted average of within-occupation conditional probabilities. Collecting terms by occupation and substituting $s^c_{j|o} = E^c_{oj}/E^c_o$ yields the second equality, where

[7] Because the 1990 state sample identifies geography only at the PUMA level and suppresses metropolitan area information, we follow Albouy et al. (2016) and assign counties and CBSAs to PUMAs using the MABLE98/Geocorr1990 crosswalk provided by the Missouri Census Data Center (`https://mcdc.missouri.edu/applications/geocorr1990.html`).

[8] A closely related measure of skill transferability is proposed by Yi et al. (2024), who infer cross-sector economic distance from wage changes of exogenously displaced workers who switch sectors. While their empirical design carefully addresses selection into switching, the resulting wage-based measure captures the *price* of reallocation conditional on switching. By contrast, our probability-based measure of skill portability captures the *feasibility* of within-occupation reallocation implied by local industry structure. Moreover, wage changes upon switching may still reflect contemporaneous demand conditions in destination sectors and therefore need not map one-to-one into underlying skill compatibility. The trade-off of our approach is that it abstracts from wage outcomes and treats human capital as predominantly occupation-specific, an assumption supported by evidence that occupations account for substantially more variation in worker outcomes than sectors (e.g. Kambourov and Manovskii (2009); Traiberman (2019)). Nevertheless, other strands of the literature emphasize the role of industry-specific human capital (e.g. Neal (1995); Parent (2000)).

skill portability is expressed as an occupation-share-weighted measure of local cross-industry dispersion, $D_o^c = 1 - \sum_j \left(s_{j|o}^c\right)^2$, which increases when a given occupation is more evenly distributed across industries.

Intuitively, P_c captures a city's reallocation flexibility—the likelihood that workers can shift across industries without changing occupations when an industry-specific shock occurs. Occupations concentrated in a single industry (e.g., *airline pilots*, *steelworkers*) rely on highly specific skills and contribute little to P_c, whereas occupations spanning multiple industries (e.g., *accountants*, *software engineers*, *sales managers*) embody more general, portable skills and contribute more. By construction, P_c is related to a city's industrial diversity but emphasizes its occupational and human-capital dimension, capturing how broadly each occupation's skills are deployed across industries. As a probability-based measure, P_c also reflects the scale advantage of larger cities, which tend to host a more complete set of occupations within each industry. Cities with higher P_c are therefore expected to adjust more easily to adverse industry shocks, as workers can be reallocated across industries without necessarily changing occupations. As shown in Fig. 6, skill portability lies predominantly between 0.55 and 0.8 and is higher in larger cities, reflecting their broader occupational presence across industries, but increases with city size at a diminishing rate as occupation–industry coverage approaches completeness.

Figure 6: Skill portability and city size. Skill portability is constructed from the 1990 5% IPUMS sample as the occupation-share-weighted probability of reallocating across local industries while remaining in the same occupation (see equation 13). Cities (CBSAs) are grouped into 100 quantiles by log employment; points show within-bin means.

4.3 Establishments

The U.S. County Business Patterns (CBP) dataset provides county–industry information on the number of establishments by employment size class.[9] Using the 1990 CBP data, this paper constructs city-level measures of total establishments and the prevalence of medium-to-large establishments. The latter is measured as the share of city employment accounted for by establishments with more than 100 employees, which proxies for the presence of larger production units that may have greater organizational and adjustment capacity. Because establishment-level employment is not directly

[9]Publicly available at `https://www.census.gov/programs-surveys/cbp/data/datasets.html`.

observed, employment for each establishment is approximated using the midpoint of its employment size class (e.g., 375 for the 250–499 category).

4.4 PNTR exposure and related controls

The PNTR exposure data are drawn from Pierce and Schott (2020), and the empirical implementation follows Pierce and Schott (2016, 2020); Pierce et al. (2024). Exposure to the granting of permanent normal trade relations (PNTR) to China is measured using industry-level *NTR gaps*, defined as the difference between non-NTR tariff rates and the NTR rates applied to WTO members for each SIC industry j:

$$NTRGap_j = NonNTRRate_j - NTRRate_j.$$

Prior to 2000, China's access to NTR tariffs was temporary and subject to annual congressional renewal, creating policy uncertainty for US–China trade. The passage of PNTR in 2000, ahead of China's accession to the WTO in 2001, eliminated this uncertainty by permanently locking in low tariff rates and generated an industry-specific trade liberalization concentrated in manufacturing. Industries with larger NTR gaps therefore experienced larger reductions in trade-policy risk following PNTR and greater subsequent exposure to the China trade shock. As shown by Pierce and Schott (2016), most cross-industry variation in the NTR gap is driven by historically predetermined non-NTR tariff rates, leaving little scope for reverse causality from contemporaneous industry conditions. CBSA-level PNTR exposure is constructed by aggregating industry-level NTR gaps using pre-policy (1990) employment shares:

$$PNTRexp_c = \sum_j \frac{L_{jc}^{1990}}{L_c^{1990}} NTRGap_j,$$

where c indexes CBSAs, j indexes industries, and L_{jc}^{1990}/L_c^{1990} denotes the share of CBSA c's total employment in industry j in 1990. As shown in Figure 7, the distribution of $PNTRexp_c$ has a mean of 7.15 percent and a standard deviation of 4.98 percent, with an interquartile range of 6.29 percentage points.

Following Pierce and Schott (2020), the baseline specification controls for contemporaneous policy changes and demographic characteristics. Two time-varying controls are included. First, we control for the average U.S. import tariff faced by each CBSA, constructed as the employment-share-weighted tariff across its active industries. Second, we account for local exposure to the removal of quantitative restrictions on textile and apparel imports associated with the phase-out of the Multi-Fiber Arrangement (MFA).

We allow for time-varying effects of several CBSA characteristics fixed at their 1990 levels: (i) the homeownership rate, to account for housing lock-in and mobility frictions; (ii) the initial unemployment rate, to proxy for persistent labour-market slack; (iii) log average household income, to control for baseline affluence and wealth-based resilience; and (iv) the foreign-born population share, to account for immigration-related labour supply shifts. Finally, we control for cities' exposure to changes in Chinese trade policy following WTO accession, measured using Chinese import tariffs on U.S. goods and Chinese domestic production subsidies.

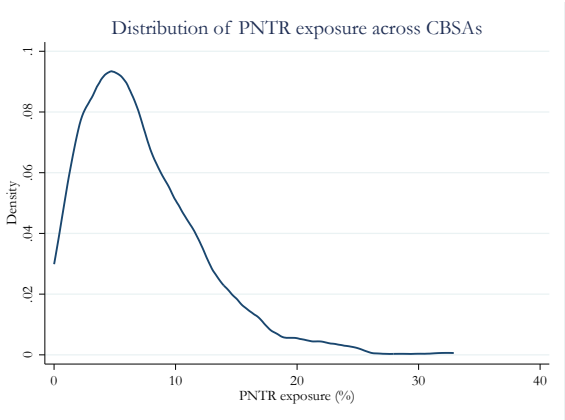


Figure 7: Kernel density of PNTR exposure across CBSAs. PNTR exposure is constructed as the employment-share-weighted average of industry-level NTR gaps using 1990 employment shares. Data is based on Pierce and Schott (2020).

5 CHINA TRADE SHOCK AND WAGE DYNAMICS

The preceding sections show that large cities' growth–risk advantage is not fully explained by inherited industry composition, but is concentrated in margins related to within-industry shock transmission and across-industry labour reallocation. To examine these mechanisms more directly, this section studies how cities respond to a large adverse industry shock: U.S. trade liberalization with China following the granting of Permanent Normal Trade Relations (PNTR). This shock is useful because its timing is well defined and its intensity varies across cities through predetermined differences in initial industry composition. We focus on two predetermined city characteristics that map naturally to the adjustment margins highlighted above: skill portability, capturing the ease of reallocating workers across industries, and the prevalence of medium-to-large establishments, capturing differences in within-industry shock transmission. This setting allows us to ask whether cities with stronger reallocation capacity and more muted shock transmission experience smaller wage losses.

5.1 Background and Event-Study Design

The China trade shock. Before 2000, China's access to U.S. Normal Trade Relations (NTR) tariff rates was temporary and subject to annual congressional renewal. Failure to renew NTR would have subjected Chinese goods to the much higher non-NTR tariffs set under the Smoot–Hawley Tariff Act of 1930, creating uncertainty over firms' long-run sourcing from China. The U.S. passage of PNTR in October 2000, shortly before China's accession to the WTO in 2001, permanently locked in NTR tariff treatment and reduced trade-policy uncertainty. Since the size of this uncertainty reduction varied across industries through predetermined NTR gaps, cities with greater initial employment in high-gap industries experienced larger exposure to the China trade shock. Details on the construction of industry-level NTR gaps and city-level PNTR exposure are provided in Section 4.

Prior work shows that PNTR generated large and spatially heterogeneous effects on U.S. labour markets. Handley and Limão (2017) estimate that the reduction in trade-policy uncertainty associ-

ated with PNTR accounts for 22%–30% of post-WTO U.S. import growth from China. Subsequent work documents adverse effects on employment, labour-force participation, and earnings in more exposed regions and workers (Pierce and Schott, 2016, 2020; Pierce et al., 2024; Rodríguez-Clare et al., 2026). Most closely related to this paper, Pierce and Schott (2020) show that the earnings effects of PNTR operate primarily through workers' local labour market exposure rather than only through their own industry, underscoring the importance of local adjustment and industry structure for shock propagation.

Outcome. The event-study specification uses log city wages as the outcome. Because log wages cumulate past wage growth,

$$\ln w_{ct} = \ln w_{c,t_0} + \sum_{\tau=t_0+1}^{t} g_{c\tau},$$

the estimated coefficients trace cumulative deviations in city wage paths rather than transitory one-period responses. This is appropriate for studying PNTR because trade liberalization affects local wages gradually through initial industry exposure, labour reallocation, and within-industry wage adjustment. From the portfolio perspective, the outcome captures the cumulative transmission of an adverse industry shock into the city-wide wage path.

5.2 Identification Strategy

This section examines the dynamic wage effects of PNTR exposure and tests whether predetermined city characteristics—skill portability and the prevalence of medium-to-large establishments—moderate the transmission of this shock. We estimate the following dynamic difference-in-differences specification:

$$\begin{aligned} \ln w_{ct} = & \sum_{t\neq 1999} \theta_t \, \mathbf{1}\{year = t\} \times PNTRexp_c \; + \sum_{t\neq 1999} \mathbf{1}\{year = t\} \times PNTRexp_c \times Z_c'\Gamma_t \\ & + \; X_{ct}\beta \; + \sum_{t\neq 1999} \lambda_t \, \mathbf{1}\{year = t\} \times X_c \; + \; \delta_c + \delta_t + \varepsilon_{ct}, \end{aligned} \tag{14}$$

where $\ln w_{ct}$ is the average real wage in city (CBSA) c in year $t \in [1990, 2013] \setminus \{1999\}$, with 1999 serving as the baseline year prior to the policy change; $\mathbf{1}\{year = t\}$ denotes year indicators; $PNTRexp_c$ captures city c's exposure to Chinese import competition induced by the resolution of trade policy uncertainty in 2000, constructed using its 1990 industry employment shares.

The vector Z_c contains predetermined, time-invariant city characteristics that moderate the impact of the trade shock. Its composition changes across the empirical exercises. First, to establish the baseline buffering role of urban scale, Z_c consists only of initial city size, $\ln Emp_c$. Second, to examine the candidate mechanisms, $Z_c = (SP_c, EQ_c)'$, where SP_c denotes skill portability and EQ_c denotes the prevalence of medium-to-large establishments. Finally, for the mediation analysis, $Z_c = (\ln Emp_c, SP_c, EQ_c)'$, allowing the mechanism variables and city size to enter jointly.

The coefficients θ_t trace the dynamic wage response to PNTR exposure for an average city, capturing the effect of inherited industry exposure to the trade shock. Since the moderator variables in Z_c are demeaned in the estimation, θ_t can be interpreted as the PNTR-induced wage path for a city

with average moderator values. The coefficient vector Γ_t captures heterogeneity in this response with respect to the included city characteristics, corresponding to the within-industry shock-transmission and across-industry reallocation margins highlighted by the decomposition. Formally, the marginal effect of PNTR exposure on city wages in year t is

$$\frac{\partial \ln w_{ct}}{\partial PNTRexp_c} = \theta_t + Z_c' \Gamma_t.$$

To identify the causal impact of PNTR exposure while assessing the role of local adjustment capacities, the specification flexibly controls for a host of potential confounders. X_{ct} includes time-varying controls for tariff exposure and exposure to the MFA phase-out. Furthermore, X_c collects a set of initial 1990 city attributes—including the mediator variables, the initial unemployment rate, homeownership rate, average household income, and the share of foreign-born population—as well as exposure to contemporaneous Chinese trade policy, including changes in Chinese tariffs and domestic production subsidies. Allowing these initial characteristics to have flexible year-specific effects mitigates concerns that other shocks or secular trends affecting particular types of cities are spuriously attributed to the PNTR response. δ_c and δ_t denote city and year fixed effects, respectively. Standard errors are clustered at the CBSA level.

5.3 Baseline Results: City Size and Shock Absorption

This section examines the dynamic wage effects of PNTR exposure and its heterogeneity by initial city size using Equation 14. We set the moderator Z_c to demeaned 1990 log employment ($\ln Emp_c$) to capture the buffering (or amplifying) role of urban scale, while also including $\ln Emp_c$ in the control set X_c to absorb secular size-dependent wage trends. Because Z_c is demeaned, the baseline coefficients θ_t isolate the dynamic PNTR effect for an average-sized city, whereas the interaction coefficients $\Gamma_t = \gamma_{0,t}$ capture the marginal effect of city scale on shock absorption. To transparently convey economic magnitudes, we plot the coefficients scaled by their respective interquartile ranges (IQRs). The plotted baseline effect, $\tilde{\theta}_t = \theta_t \times IQR^{PNTR}$, translates to the implied wage impact of an IQR increase in PNTR exposure on an average-sized city. The plotted interaction effect applies a double-IQR scaling, $\tilde{\gamma}_{0,t} = \gamma_{0,t} \times IQR^{PNTR} \times IQR^{\ln Emp}$, to quantify the relative buffering capacity. It measures the difference in wage trajectories between a 75th- and 25th-percentile city facing an identical IQR shift in trade shock exposure.

Given the large number of estimated parameters, Figure 8 plots the dynamic estimates.[10] To rule out confounding from pre-existing local wage dynamics, we present both a baseline specification (Panels a and b) and a detrended model that partials out city-specific pre-1999 linear trends (Panels c and d).[11] Panels (a) and (c) display the scaled baseline event-study coefficients ($\tilde{\theta}_t$), capturing the wage impact of an interquartile (IQR) shift in PNTR exposure. There is no evidence of differential pre-trends prior to 2000 once city-specific pre-trends are accounted for. Following China's WTO accession, wages fall significantly in more exposed cities. Focusing on the baseline results in Panel (a), this negative effect intensifies through the late 2000s and reaches approximately -0.037 log points by 2010. To interpret this, an IQR increase in PNTR exposure translates to a 3.6% wage

[10] Full regression tables are available upon request.

[11] See Appendix E.2 for details of the de-trending procedure. The rest of the main paper focuses on the detrended specification, while results based on the undetrended specification are reported in the Appendix.

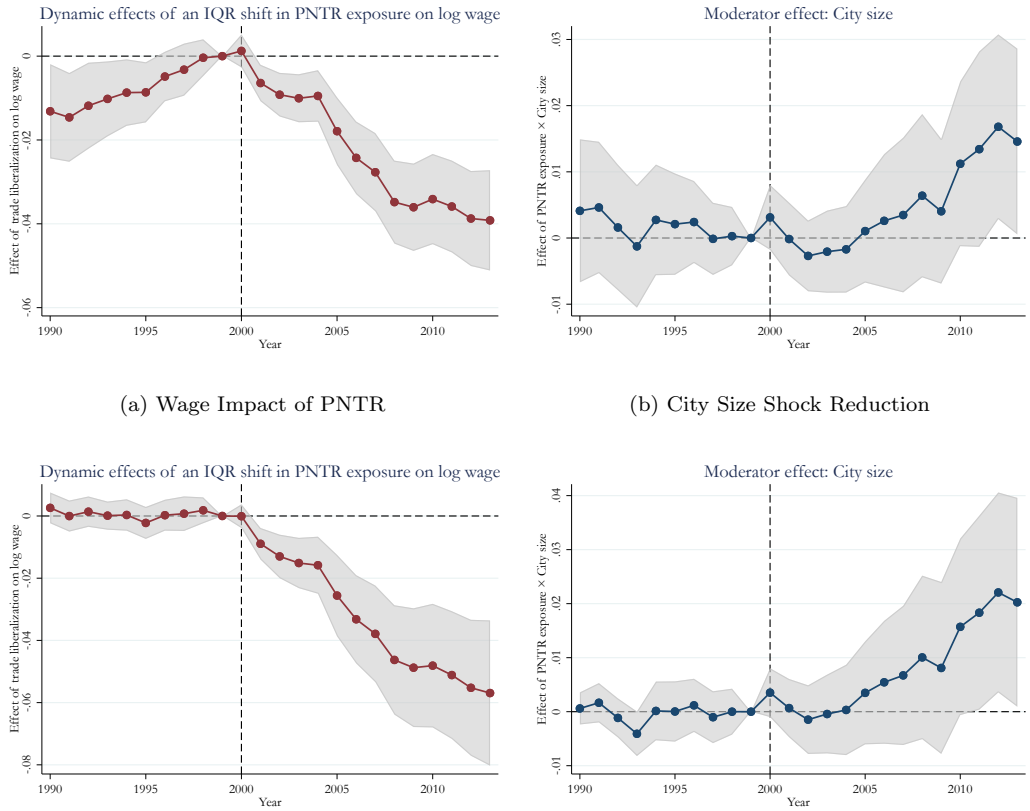
decline for an average-sized city ($e^{-0.037} - 1 \approx -3.6\%$).[12]

Panels (b) and (d) report the scaled triple-interaction coefficients ($\tilde{\gamma}_{0,t}$), capturing heterogeneity in the shock's impact by city size. The estimates turn positive and statistically significant post-2000, suggesting that larger cities attenuated local wage losses. Focusing on the baseline results in Panel (b), this buffering effect reaches approximately 0.01 log points by 2010. To interpret this, moving from the 25th to the 75th percentile of the city size distribution averts a 1% wage drop when facing an identical IQR trade shock ($e^{0.010} - 1 \approx 1.0\%$). Remarkably, this spatial advantage offsets roughly 27% of the baseline wage decline (i.e., 0.01/0.037).[13]

Overall, the negative wage effects of geographic-level China shock exposure align with standard trade models and empirical evidence. For example, consistent with models where import competition under trade imbalances depresses local labour demand and transmits beyond manufacturing through reduced local expenditure, Autor et al. (2013) find that a $1,000 per worker increase in a Commuting Zone's trade exposure reduced mean weekly earnings by roughly 0.76% over a decade. Using matched employer–employee data, Pierce et al. (2024) document persistent earnings losses for manufacturing workers following PNTR and find that own-county exposure dominates own-industry exposure, even when accounting for input–output linkages. However, relatively little work has examined how local labour-market characteristics beyond industry composition shape how the China shock translates into heterogeneous wage responses. A notable exception is Bloom et al. (2024); while focusing on employment reallocation rather than wages, they document that large high-human-capital coastal labour markets experience net employment gains of about 0.5% per year as service expansion offsets manufacturing losses. Building on this, our results establish city size as a first-order margin of spatial resilience, prompting our subsequent investigation into the mechanisms underlying this scale-based shock absorption.

[12]Following $\tilde{\theta}_t = \theta_t \times IQR^{PNTR}$, the plotted 2010 value corresponds to the raw point estimate ($\theta_{2010} \approx -0.006$) multiplied by the 6.29 percentage-point IQR in PNTR exposure.

[13]Following $\tilde{\gamma}_{0,t} = \gamma_{0,t} \times IQR^{PNTR} \times IQR^{\ln Emp}$, the plotted 2010 value $\tilde{\gamma}_{0,2010} = 0.010$ corresponds to the raw point estimate ($\gamma_{0,2010} \approx 0.001$) multiplied by the respective IQRs of PNTR exposure (6.29 percentage points) and initial city log employment (1.59 log points). To give an alternative sense of the magnitude, a doubling of city size ($\Delta \ln Emp \approx 0.69$) attenuates the wage effect of an IQR trade shock by about 0.004 log points ($0.001 \times 0.69 \times 6.29 \approx 0.004$), offsetting roughly 11% of the baseline decline.



(a) Wage Impact of PNTR

(b) City Size Shock Reduction

(c) Wage Impact of PNTR (Detrended)

(d) City Size Shock Reduction (Detrended)

Figure 8: Dynamic wage effects of an IQR shift in PNTR exposure and heterogeneity in effects by city size. Panels (a) and (c) plot the scaled baseline event-study coefficients $\tilde{\theta}_t$ capturing the dynamic effect of an interquartile range (IQR) shift in PNTR exposure on city-level log wages (relative to 1999). Panels (b) and (d) plot the scaled triple-interaction coefficients $\tilde{\gamma}_{0,t}$, which capture the added buffering effect of an IQR shift in initial city size (1990 log employment) against the same trade shock. Panels (a) and (b) report estimates from the baseline specification. Panels (c) and (d) report estimates from an alternative specification that partials out city-specific pre-existing linear wage trends (estimated using pre-1999 data). All specifications include city and year fixed effects and flexibly control for time-varying effects of initial city attributes as in Equation 14. Shaded areas denote 95% confidence intervals, with standard errors clustered at the CBSA level. Data from QCEW and Pierce and Schott (2020).

5.4 Mechanisms of Urban Shock Absorption

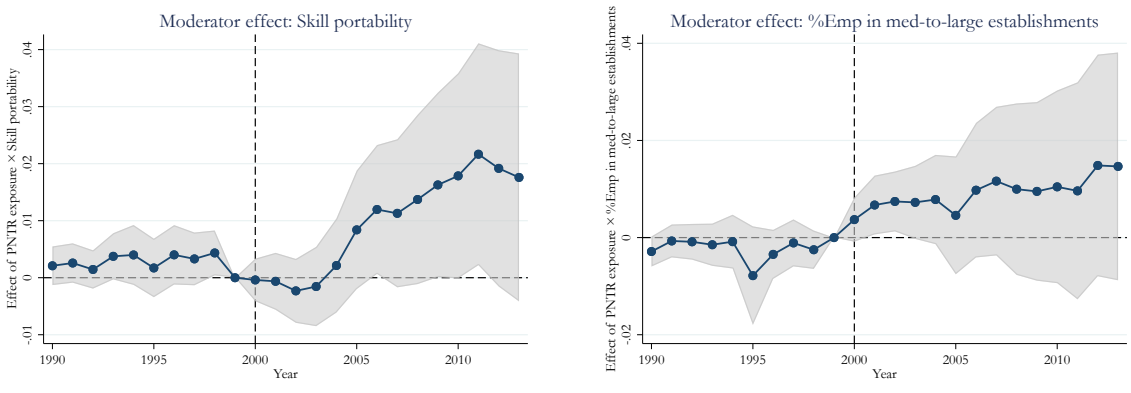
Having established that large cities buffer the China shock, we now examine two local adjustment channels suggested by the decomposition: skill portability and the prevalence of medium-to-large establishments. Skill portability captures the worker-side ability to reallocate labour across industries after sectoral disruptions. The prevalence of medium-to-large establishments captures a complementary within-industry margin, reflecting differences in the organizational scale through which shocks may be absorbed before being transmitted to local wages. We use County Business Patterns (CBP) as the full-sample baseline and supplement this analysis with Compustat evidence on publicly listed firms in Appendix E.1.[14]

To evaluate these channels empirically, we set the moderator vector to $Z_c' = (SP_c,\ EQ_c)'$ in Equation 14. Here, SP_c denotes pre-shock city-level skill portability, measured in 1990 as the expected probability that a worker can reallocate to another local industry while remaining in the same occupation. EQ_c denotes city c's 1990 employment share in establishments with more than 100 employees. Figure 9 plots the double-IQR scaled coefficients, $\tilde{\gamma}_{1,t}$ and $\tilde{\gamma}_{2,t}$, which measure the difference in wage trajectories between cities at the 75th and 25th percentiles of each mechanism when facing the same interquartile shift in PNTR exposure.

Using the detrended wage specification, Panel (a) shows that the moderating effect of skill portability is flat before PNTR, then turns positive and statistically significant after 2005. By 2010, the effect reaches approximately 0.018 log points, implying that moving from the 25th to the 75th percentile of the skill-portability distribution averts a 1.8% wage decline ($e^{0.018} - 1 \approx 1.8\%$) and offsets roughly 49% of the baseline aggregate wage loss (0.018/0.037). Panel (b) shows that the prevalence of medium-to-large establishments also exhibits no differential pre-trend and becomes positive after China's WTO accession. The estimates are statistically significant mainly in the short run, plausibly reflecting measurement imprecision from CBP size-class data. By 2010, the buffering effect reaches about 0.01 log points, preserving roughly 1% of local wages and offsetting approximately 27% of the baseline shock (0.01/0.037).

These results suggest that urban industry shock absorption operates through both worker-side and establishment-side channels. On the across-industry margin, skill portability makes local labour supply more adaptable: displaced workers can move across industries with less skill depreciation, reducing downward wage pressure and limiting local demand spillovers. This interpretation is consistent with Bloom et al. (2024), who document stronger post-shock reallocation in high-human-capital labour markets. On the within-industry margin, the buffering role of medium-to-large establishments is consistent with Kopytov et al. (2025), who emphasize the scale advantage of larger firms in internal risk management. The CBP evidence shows that cities with a higher employment share in medium-to-large establishments experience smaller wage losses after the industry shock. However, it does not reveal why larger establishments buffer shocks more effectively. Appendix E.1 provides complementary evidence from Compustat, examining whether this establishment-size margin is related to observable financial and operational characteristics of larger firms.

[14] While Compustat provides rich data on corporate financial fundamentals, its restriction to publicly listed firms results in zero observations for some smaller micropolitan areas (μSAs). This geographic limitation restricts its use to auxiliary mechanism analysis rather than the full-sample baseline.



(a) Skill Portability Buffering

(b) Med-to-large Establishment Buffering

Figure 9: The structural mechanisms of spatial resilience: Skill Portability (SP) and Establishment Quality (EQ). This figure plots the dynamic buffering effects of a city's initial skill network and firm quality against the PNTR wage shock. The estimates are obtained jointly from the baseline event-study specification (Equation 14), allowing SP and EQ to compete in a "horse race." Panel (a) plots the scaled interaction coefficients for Skill Portability ($\tilde{\gamma}_{1,t} = \gamma_{1,t} \times IQR^{PNTR} \times IQR^{SP}$), capturing the differential wage trajectory between a high-SP city (75th percentile) and a low-SP city (25th percentile) confronting an identical IQR trade shock. Panel (b) plots the analogous scaled interaction coefficients for Establishment Quality ($\tilde{\gamma}_{2,t} = \gamma_{2,t} \times IQR^{PNTR} \times IQR^{EQ}$). Both panels report estimates from a specification that partials out city-specific pre-1999 linear wage trends. All specifications include city and year fixed effects, along with flexible controls for time-varying effects of initial city attributes, including city size ($\ln Emp_c$). Shaded areas denote 95% confidence intervals, with standard errors clustered at the CBSA level.

5.5 Mediation Analysis

This section examines whether skill portability and the prevalence of medium-to-large establishments account for the scale-dependent buffering effect documented in Section 5.3. We compare the baseline city-size interaction with an augmented specification that jointly includes the interactions of PNTR exposure with skill portability, establishment structure, and city size. Specifically, the moderator vector in Equation 14 is set to

$$Z_c' = (SP_c,\ EQ_c,\ \ln Emp_c)'.$$

This horse-race specification asks how much of the city-size buffering effect remains once the two candidate adjustment margins are allowed to absorb variation in the PNTR response.

Figure 10 shows that the coefficient on the PNTR–city-size interaction declines sharply and loses statistical significance after controlling for SP_c and EQ_c. Quantitatively, the city-size interaction falls by more than 75% over a decade, indicating that these two margins account for most of the scale-dependent wage resilience documented above. This attenuation suggests that the resilience advantage of larger cities operates largely through labour-market reallocation flexibility and establishment-based shock buffering, rather than through population size per se. The evidence therefore supports the interpretation that urban resilience reflects local adjustment capacity embedded in the city's industry portfolio. At the same time, SP_c and EQ_c may also be correlated with other local attributes that shape shock absorption, such as human capital composition or access to external finance. The mediation exercise should therefore be interpreted as evidence that these two margins capture important dimensions of scale-related adjustment capacity, rather than as a claim that they exhaust all mechanisms behind the city-size effect.

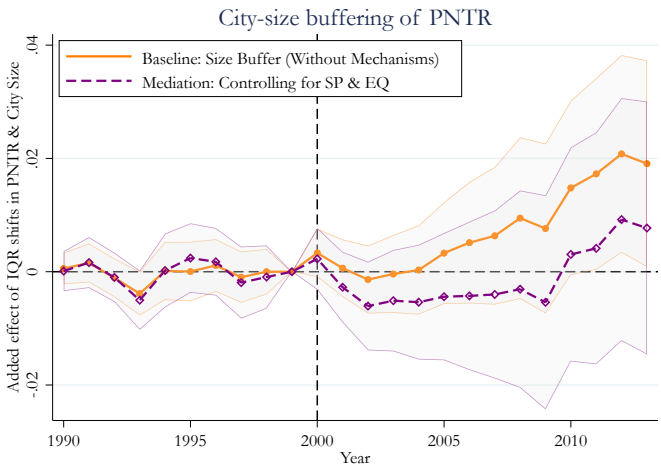


Figure 10: Mediation Analysis on the City-Size Buffering Effect. This figure plots the estimated coefficients of the triple interaction between PNTR exposure, year dummies, and city size. The solid orange line represents the baseline specification without mediators. The dashed purple line shows the remaining size effect after controlling for the interactions of PNTR with skill portability (SP) and establishment quality (EQ) as proxied by the share of city employment in establishments with above 100 employees. All coefficients are scaled to represent an IQR shift in both PNTR, city size and an respective mechanism. Shaded areas denote 95% confidence intervals. Data is from CBP, QCEW, Pierce and Schott (2020).

6 CONCLUSION

This paper advances a portfolio view of cities in which urban wage growth and wage risk are jointly shaped by inherited industrial structure and local adjustment capacity. Initial industry composition determines cities' passive exposure to nationally fluctuating industries and accounts for much of long-run wage growth. Yet it does not fully explain why some cities occupy more favourable positions in growth–risk space. The decomposition shows that large cities' advantage also operates through local adjustment margins: stronger reallocation gains on the growth side, and lower risk contributions from benchmark-relative within-industry deviations and wage-tier sorting on the risk side. Using U.S. trade liberalization with China as a spatially varying adverse shock, the paper further shows that skill portability and the prevalence of medium-to-large establishments substantially attenuate local wage losses and account for much of the scale-dependent resilience advantage. Cities are therefore not mechanically locked into their initial portfolio positions: inherited industry structure shapes exposure, but local adjustment capacity governs how shocks are transmitted once they occur.

The results point to a broader agenda. Skill portability and establishment scale are unlikely to exhaust the relevant margins of local adjustment; financial depth, labour-market institutions, housing-market flexibility, and governance capacity may also shape urban resilience. The portfolio framework could also be embedded in a dynamic spatial general equilibrium model with heterogeneous workers and firms, mobility frictions, and location-specific adjustment technologies, allowing explicit welfare analysis of cities' growth–risk trade-offs and the effects of place-based policies. More broadly, cities are not only portfolios of industries, but also bundles of housing, amenities, and local risks. A portfolio perspective therefore offers a useful language for studying location choice and policy under multiple, interacting sources of risk and return.

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